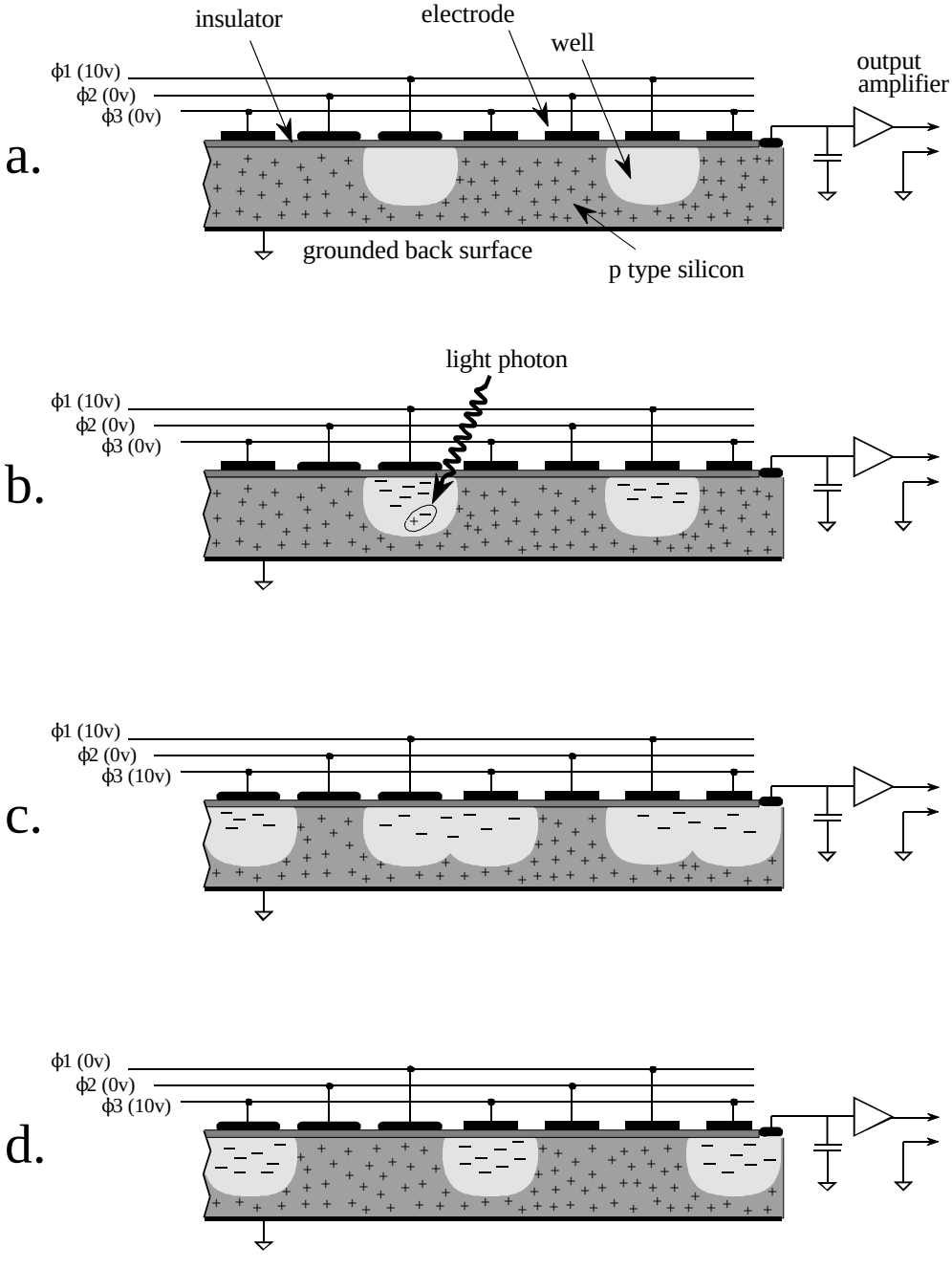

FIGURE 23-7
Operation of the charge coupled device (CCD). As shown in this cross-sectional view, a thin sheet of p-type silicon is covered with an insulating layer and an array of electrodes. The electrodes are connected in groups of three, allowing three separate voltages to be applied: ϕ1, ϕ2, and ϕ3. When a positive voltage is applied to an electrode, the holes (i.e., the positive charge carriers indicated by the "+") are pushed away. This results in an area depleted of holes, called a *well*. Incoming light generates holes and electrons, resulting in an accumulation of electrons confined to each well (indicated by the "-"). By manipulating the three electrode voltages, the electrons in each well can be moved to the edge of the silicon where a charge sensitive amplifier converts the charge into a voltage.